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MAE 3240

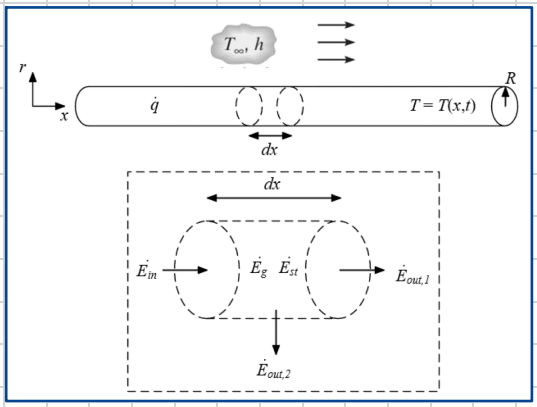
HW3

Problem 1

Area(A) = πR^2 Perimeter(P) = $2\pi R$ $\quad P = \frac{2A}{R}$

(a) $\dot{E}_{st} = \dot{E}_g + \dot{E}_{in} - \dot{E}_{out_1} - \dot{E}_{out_2}$

$$\boxed{\rho c_p A \frac{dT}{dt} dx = \dot{q} A dx + \left[-kA\frac{\partial T}{\partial x}\right]_x - \left[-kA\frac{\partial T}{\partial x}\right]_{x+dx} - \frac{2A}{R} h dx (T - T_\infty)}$$



(b) $\left[-kA\frac{\partial T}{\partial x}\right]_{x+dx} = \left[-kA\frac{\partial T}{\partial x}\right]_x + kA\frac{\partial^2 T}{\partial x^2}dx$

$$\rho c_p A \frac{dT}{dt} dx = \dot{q} A dx + \left[-kA\frac{\partial T}{\partial x}\right]_x - \left[-kA\frac{\partial T}{\partial x}\right]_{x+dx} - \frac{2A}{R} h dx (T - T_\infty)$$

$$\rho c_p A \frac{dT}{dt} dx = \dot{q} A dx + kA\frac{\partial^2 T}{\partial x^2}dx - \frac{2A}{R} h dx (T - T_\infty)$$

$$\boxed{\rho c_p \frac{dT}{dt} = \dot{q} + k\frac{\partial^2 T}{\partial x^2} - \frac{2h}{R}(T - T_\infty)}$$

(c) At $x = 0$: $\boxed{T(0,t) = T_i}$

At $x = L$: $\boxed{-k\frac{\partial T}{\partial x}\Big|_{x=L} = h(T(L,t) - T_\infty)}$

} boundary conditions

initial condition: $\boxed{T(x,0) = T_i}$

Problem 2

(a) $R_{cond,B} = \frac{L_B}{k_B} = \frac{0.1}{30} = \boxed{3.33 \times 10^{-3}}$

$R_{conv} = \frac{1}{h} = \boxed{2.0 \times 10^{-3}}$

$q'' = \frac{T_i - T_\infty}{R_{cond,B} + R_{conv}} = \frac{100°C - 20°C}{5.33 \times 10^{-3}} = \boxed{1.5 \times 10^4\ W/m^2}$

$T_s = T_\infty + q'' R_{conv} = 20°C + (1.5 \times 10^4\ W/m^2)(2.0 \times 10^{-3})$

$\boxed{T_s = 50°C}$

(b) $\dot{q} L_A = q'' \Rightarrow \dot{q} = q''/L_A$

$\dot{q} = \frac{1.5 \times 10^4\ W/m^2}{0.30\ m} = \boxed{5.0 \times 10^4\ W/m^3}$

(c) $T(x) = T_{max} - \frac{\dot{q}}{2k_A}x^2$

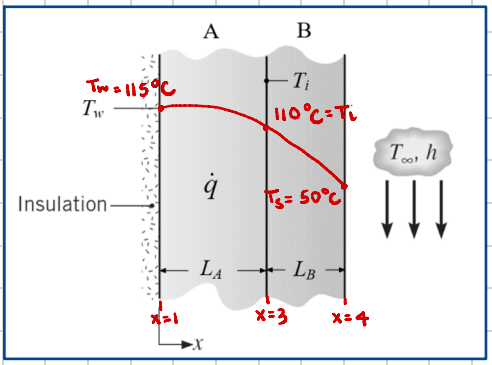
$T_i = T_w - \frac{\dot{q}}{2k_A}L_A^2$

$k_A = \frac{\dot{q} L_A^2}{2(T_w - T_i)} = \frac{(5.0 \times 10^4\ W/m^3)(0.3)^2}{2(115°C - 110°C)} = \boxed{150\ W/(m \cdot K)}$

(d) $T_w = 115°C$

$T_i = 110°C$

$T_s = 50°C$



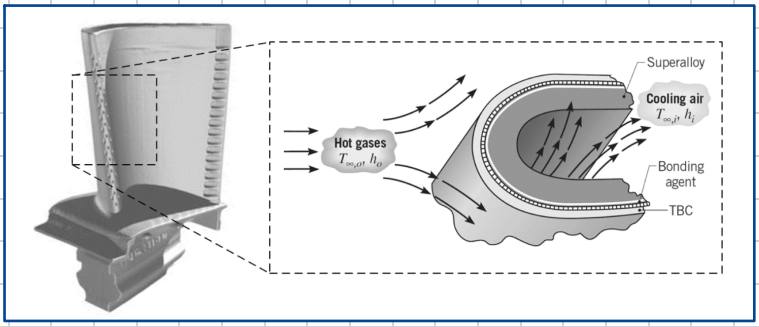
Problem 3

(a) $R_{total} = \frac{1}{h_o} + \frac{L_b}{k_b} + \frac{1}{h_i} = 0.001 + 0.0002 + 0.002 = 0.0032$

$q'' = \frac{T_{\infty,o} - T_{\infty,i}}{R_{total}} = \frac{1616K - 400K}{0.0032} = 3.8 \times 10^5$

$T_s = 1616K - q''\left(\frac{1}{h_o}\right) = \boxed{1236K}$

$\boxed{\text{No,}}$ the superalloy blade exceed T_{max}.



(b) $R_{TBC} = 0.0005$

$R_{total} = 0.001 + 0.0005 + 0.0001 + 0.0002 + 0.002 = 0.0038$

$q'' = \frac{T_{\infty,o} - T_{\infty,i}}{R_{total}} = \frac{1616K - 400K}{0.0038} = 3.2 \times 10^5$

$T_s = 1616K - q''(0.001 + 0.0005 + 0.0001) = \boxed{1100K}$

$\boxed{\text{Yes,}}$ the superalloy blade can be maintained below T_{max}.